

untangle::actuator

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Chapter 1

actuator

An actuator is a generic callable that may invoke a list of actions(callables of type `std::function<...>`).

Its goal is to provide a light and simple mechanism to connect components that want to communicate with each other. It works similar as the "signal and slots" mechanisms from other frameworks, but much simplified.

Modern programming languages do not offer class intrinsic interfaces that would improve significantly how objects are connected and how they communicate. Instead, languages, like C++, are using external interfaces, that are not only cluttering the code, but they don't feel "natural" as they are opposed to what an interface is in reality. Basically the objects and interfaces are not separated, and the interface is intrinsic to the object(think on the cogs in a mechanism). A generic callable is an attempt to mitigate this problem. `async` implementation is such use case.

An actuator provides the results of all actions in a vector, in the same order as the actions were added.

Example: how to use actuator instead of polymorphism

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle)
{
    for (const auto& s : shapes)
    {
        s->rotate(angle);
    }
}

std::shared_ptr<triangle> t(new triangle);
std::shared_ptr<circle> c(new circle);
std::shared_ptr<square> s(new square);

// using polymorphism
std::vector<shape*> shapes;
shapes.push_back(t.get());
shapes.push_back(c.get());
shapes.push_back(s.get());

rotate_shapes(shapes, 10);

// using actuator
auto action1 = untangle::bind(t, &triangle::rotate);
auto action2 = untangle::bind(c, &circle::rotate);
auto action3 = untangle::bind(s, &square::rotate);
auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

For convenience there are provided helpers methods to "connect" to an initial list of "actions", or to create bindings to class methods.

Please check the manual in [doc/refman.pdf](#) for further references.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions [11](#)

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	Interface to untangle::actuator functor	15
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

untangle::actuator< action_t >	
An actuator is a functor that can trigger a dynamic list of actions (of type std::↵ function<...>)	17
untangle::invalid_action	
Invalid action exception	22

Chapter 5

File Index

5.1 File List

Here is a list of all documented files with brief descriptions:

actuator.hpp	25
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Chapter 6

Topic Documentation

6.1 namespace untangle: functions

Functions

- `template<typename action_t, typename ... Actions>`
`actuator< action_t > untangle::connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t untangle::bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t untangle::bind (class_t *obj, T class_t::*method)`
Binding to a class method.

6.1.1 Detailed Description

6.1.2 Function Documentation

6.1.2.1 connect()

```
template<typename action_t, typename ... Actions>
actuator< action_t > untangle::connect (
    action_t & A1,
    Actions &... An)
```

Creates an actuator holding an initial list of actions.

Parameters

A1	- The first action. It is specified as <code>std::function<...></code> .
----	--

An	- Any number of further actions, of the same type as A1.
----	--

Returns

An [actuator](#).

Example:

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle)
{
    for (const auto& s : shapes)
    {
        s->rotate(angle);
    }
}

auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

6.1.2.2 bind() [1/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::<
type>>>
action_t untangle::bind (
    const std::shared_ptr< class_t > & obj,
    T class_t::* method)
```

Binding to a class function member.

It returns a `std::function(lambda)` that wraps the function member. It may be used to provide an action for [connect\(\)](#) or [actuator::add\(\)](#).

Remarks

It requires a shared pointer to the class type. A `std::weak_ptr` to it is captured internally in a lambda, so the binding can always check whether the shared object is still alive without keeping it alive itself. Therefore, it is safe to use actions provided by this binding inside an [actuator](#), and safe for the action to outlive the caller's shared pointer.

Parameters

obj	- Class object.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

`action_t` - A `std::function` that wraps the pointer to function member.

Remarks

If the class object gets invalid, invoking this binding will throw an exception of type [invalid_action](#).

6.1.2.3 bind() [2/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::↵
type>>
action_t untangle::bind (
    class_t * obj,
    T class_t::* method)
```

Binding to a class method.

Attention

It is not safe to use this binding when the pointed-to object may be destroyed. A null pointer is detected and reported as a dead action, but a pointer to an already destroyed object can not be distinguished from a valid one. Prefer the overload taking a `std::shared_ptr`.

Remarks

It is provided for convenience of use: within a class it is safe to create bindings through this pointer.

Binding a null pointer produces an action that is always dead: invoking it throws [invalid_action](#), and an [actuator](#) drops it.

Parameters

obj	- Pointer to class.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

action_t - A `std::function` that wraps the pointer to function member.

Chapter 7

Namespace Documentation

7.1 untangle Namespace Reference

Interface to `untangle::actuator` functor.

Classes

- struct `invalid_action`
Invalid action exception.
- struct `actuator`
An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Functions

- `template<typename action_t, typename ... Actions>`
`actuator< action_t > connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (class_t *obj, T class_t::*method)`
Binding to a class method.

7.1.1 Detailed Description

Interface to `untangle::actuator` functor.

Chapter 8

Class Documentation

8.1 untangle::actuator< action_t > Struct Template Reference

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

```
#include <actuator.hpp>
```

Public Types

- using `actions_t` = `std::list<action_t*>`
Actions container type.
- using `results_t` = `std::vector<typename result_t::type>`
Results container type.

Public Member Functions

- `actuator & operator= (const actuator &other)=default`
Assignment operator.
- `action_t type () const`
Helper method to access the action type of this object.
- `void reset ()`
Remove all actions and any stored results.
- `template<typename ... Args>`
`void operator() (Args &&... args)`
The call operator.
- `template<typename ... Args>`
`void invoke_action (const std::string &name, Args &&... args)`
Invokes one single action associated with a key.
- `void add (action_t *action)`
Add an action to the actions list.
- `void add (const std::string &name, action_t *action)`
Add action to the actions map associated with a name.
- `void remove (const action_t *action)`
Remove an action from the actions list.
- `void remove (const std::string &name)`
Remove an action from actions map.
- `bool is_connected () const`
Check if this actuator is "connected" with other actions.
- `bool has_action (const std::string &name) const`
Check if there is certain named action.

Public Attributes

- [actions_t](#) actions
Actions list.
- [actions_map_t](#) [actions_map](#)
Named actions map.
- [results_t](#) results
Actions return values list.

8.1.1 Detailed Description

```
template<typename action_t>
struct untangle::actuator< action_t >
```

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Remarks

An actuator object can be constructed with an initial list of actions by [connect\(\)](#).

Template Parameters

<code>action_t</code>	Action type. It is specified as <code>std::function<...></code> .
-----------------------	---

8.1.2 Member Typedef Documentation

8.1.2.1 `actions_t`

```
template<typename action_t>
using untangle::actuator< action_t >::actions_t = std::list<action_t*>
```

Actions container type.

Remarks

The elements stored are of pointer type, that is required to implement the [remove\(\)](#) operation. `std::function` supports only equality operator for nullptr (two `std::function(s)` can not compare).

8.1.2.2 `results_t`

```
template<typename action_t>
using untangle::actuator< action_t >::results_t = std::vector<typename result_t::type>
```

Results container type.

It holds the return values of the actions that have a non-void return type. Upon the actuator invocation, the returns can be extracted from [results](#).

8.1.3 Member Function Documentation

8.1.3.1 operator=()

```
template<typename action_t>
actuator & untangle::actuator< action_t >::operator= (
    const actuator< action_t > & other) [default]
```

Assignment operator.

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
untangle::actuator<decltype(actuator_rotate.type())> actuator_rotate_1 = actuator_rotate;
actuator_rotate_1(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.2 type()

```
template<typename action_t>
action_t untangle::actuator< action_t >::type () const [inline]
```

Helper method to access the action type of this object.

Note

Intended to be used in expressions by `decltype(<actuator instance>.type())`

Returns

action_t

8.1.3.3 reset()

```
template<typename action_t>
void untangle::actuator< action_t >::reset () [inline]
```

Remove all actions and any stored results.

After this call the actuator is empty: `actuator::is_connected()` returns false.

8.1.3.4 operator()()

```
template<typename action_t>
template<typename ... Args>
void untangle::actuator< action_t >::operator() (
    Args &&... args) [inline]
```

Parameters

args	- Arguments list must match the action arity.
------	---

8.1.3.5 invoke_action()

```
template<typename action_t>
template<typename ... Args>
void untangle::actuator< action_t >::invoke_action (
    const std::string & name,
    Args &&... args) [inline]
```

Invokes one single action associated with a key.

Parameters

name	- Key associated with the action.
args	- Arguments list must match the action arity.

8.1.3.6 add() [1/2]

```
template<typename action_t>
void untangle::actuator< action_t >::add (
    action_t * action) [inline]
```

Add an action to the actions list.

Parameters

action	- Action to be added.
--------	-----------------------

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(2).WillRepeatedly(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.add(&action1);
actuator_rotate(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.7 add() [2/2]

```
template<typename action_t>
void untangle::actuator< action_t >::add (
```


Parameters

name	- Name of the action.
action	- Action to be added.

8.1.3.8 remove() [1/2]

```
template<typename action_t>
void untangle::actuator< action_t >::remove (
    const action_t * action) [inline]
```

Remove an action from the actions list.

An invalid action (empty std::function) is implicitly removed when `operator()()` is invoked.

Parameters

action	- Action to be removed.
--------	-------------------------

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(0);
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.remove(&action1);
actuator_rotate(50);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.9 remove() [2/2]

```
template<typename action_t>
void untangle::actuator< action_t >::remove (
    const std::string & name) [inline]
```

Remove an action from actions map.

Parameters

name	- Name of the action to remove.
------	---------------------------------

8.1.3.10 is_connected()

```
template<typename action_t>
bool untangle::actuator< action_t >::is_connected () const [inline]
```

Check if this actuator is "connected" with other actions.

Returns

- true - if the `actuator::actions` list is not empty.
- false - if the `actuator::actions` list is empty.

8.1.3.11 has_action()

```
template<typename action_t>
bool untangle::actuator< action_t >::has_action (
    const std::string & name) const [inline]
```

Check if there is certain named action.

Parameters

name	- Name associated with the action.
------	------------------------------------

Returns

- true - if name can be found in `actuator::actions_map`
- false - if name can not be found in `actuator::actions_map`

The documentation for this struct was generated from the following file:

- `actuator.hpp`

8.2 untangle::invalid_action Struct Reference

Invalid action exception.

```
#include <actuator.hpp>
```

Public Member Functions

- `invalid_action` (std::string text)
Construct a new invalid action object.
- `const char * what () const` noexcept override
The message text describing the reason of this exception.

8.2.1 Detailed Description

Invalid action exception.

Remarks

An action may be provided as a binding to a class function member, by using [untangle::bind\(\)](#). When the class object gets invalid, invoking the action will raise an exception to this type.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 invalid_action()

```
untangle::invalid_action::invalid_action (  
    std::string text) [inline], [explicit]
```

Construct a new invalid action object.

Parameters

text	- A message text, describing the reason of this exception.
------	--

8.2.3 Member Function Documentation

8.2.3.1 what()

```
const char * untangle::invalid_action::what () const [inline], [override], [noexcept]
```

The message text describing the reason of this exception.

Returns

const char* - The message text.

The documentation for this struct was generated from the following file:

- actuator.hpp

Chapter 9

File Documentation

9.1 actuator.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #pragma once
00007
00008 #include <vector>
00009 #include <list>
00010 #include <map>
00011 #include <string>
00012 #include <utility>
00013 #include <functional>
00014 #include <memory>
00015 #include <iostream>
00016 #include <type_traits>
00017 #include <exception>
00018
00019 namespace untangle
00020 {
00021 // exception
00029 struct invalid_action : std::exception
00030 {
00036 explicit invalid_action(std::string text) : message(std::move(text)){}
00037
00043 const char* what() const noexcept override { return message.c_str(); }
00044
00045 private:
00046 std::string message;
00047 };
00048
00056 template<typename action_t>
00057 struct actuator final
00058 {
00065 using actions_t = std::list<action_t*>;
00066 using actions_map_t = std::map<std::string, action_t*>;
00067 using result_t = std::conditional<std::is_void<typename action_t::result_type>::value, int, typename
    action_t::result_type>;
00074 using results_t = std::vector<typename result_t::type>;
00075
00076 actions_t actions;
00077 actions_map_t actions_map;
00078 results_t results;
00079
00080 actuator() = default;
00081 actuator(const actuator&) = default;
00082 actuator(actuator&&) noexcept = default;
00083 ~actuator() = default;
00090 actuator& operator=(const actuator& other) = default;
00091 actuator& operator=(actuator&&) noexcept = default;
00092
00100 action_t type() const
00101 {
00102     return nullptr;
00103 }
00104
00110 void reset()
00111 {
00112     actions.clear();
00113     actions_map.clear();

```

```

00114     results.clear();
00115 }
00116
00124 template<typename ...Args>
00125 void operator()(Args&&... args)
00126 {
00127     results.clear();
00128
00129     // Dead bindings are collected here and dropped after the loop.
00130     // The actuator does not own the actions it points at, so it must never
00131     // write through action_t* into a std::function belonging to the caller.
00132     std::vector<action_t*> dead_actions;
00133
00134     for (const auto& action : actions)
00135     {
00136         // A null pointer or an empty std::function can never be invoked. Calling an
00137         // empty one throws std::bad_function_call, which is not an invalid_action and
00138         // would escape this operator, so drop it instead of invoking it.
00139         if (action == nullptr || !*action)
00140         {
00141             dead_actions.push_back(action);
00142             continue;
00143         }
00144
00145         try
00146         {
00147             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00148                 (*action)(std::forward<Args>(args)...);
00149             } else {
00150                 results.push_back((*action)(std::forward<Args>(args)...));
00151             }
00152         }
00153         catch (const invalid_action& ia)
00154         {
00155             std::cout << ia.what() << std::endl;
00156             dead_actions.push_back(action);
00157         }
00158     }
00159
00160     for (const auto& dead_action : dead_actions)
00161     {
00162         actions.remove(dead_action);
00163     }
00164 }
00165
00172 template<typename ...Args>
00173 void invoke_action(const std::string& name, Args&&... args)
00174 {
00175     results.clear();
00176     const auto& it = actions_map.find(name);
00177     if (it != actions_map.end())
00178     {
00179         try
00180         {
00181             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00182                 (*it->second)(std::forward<Args>(args)...);
00183             } else {
00184                 results.push_back((*it->second)(std::forward<Args>(args)...));
00185             }
00186         }
00187         catch (const invalid_action& ia)
00188         {
00189             std::cout << ia.what() << std::endl;
00190             actions_map.erase(name);
00191         }
00192     }
00193 }
00194
00203 void add(action_t* action)
00204 {
00205     actions.push_back(action);
00206 }
00207
00214 void add(const std::string& name, action_t* action)
00215 {
00216     actions_map.emplace(name, action);
00217 }
00218
00229 void remove(const action_t* action)
00230 {
00231     actions.remove_if([&action](const auto& a)
00232     {
00233         return (action == a);
00234     });
00235 }
00236
00242 void remove(const std::string& name)

```

```

00243 {
00244     actions_map.erase(name);
00245 }
00246
00253 bool is_connected() const { return !actions.empty() || !actions_map.empty(); }
00254
00262 bool has_action(const std::string& name) const { return actions_map.find(name) != actions_map.end(); }
00263 };
00264
00279 template<typename action_t, typename ...Actions>
00280 actuator<action_t> connect(action_t& A1, Actions&... An)
00281 {
00282     using actuator_t = untangle::actuator<action_t>;
00283     actuator_t actuator;
00284     actuator.actions = {&A1, &An...};
00285
00286     // remove empty actions
00287     actuator.actions.remove_if([](const auto& action)
00288     {
00289         return (*action == nullptr);
00290     });
00291     return actuator;
00292 }
00293
00294 template <typename actuator_t>
00295 void remove_empty_actions(actuator_t& actuator)
00296 {
00297     std::vector<typename actuator_t::actions_map_t::const_iterator> removable_iterators;
00298     for (auto it = actuator.actions_map.begin(); it != actuator.actions_map.end(); ++it)
00299     {
00300         if (it->second == nullptr)
00301         {
00302             removable_iterators.push_back(it);
00303         }
00304     }
00305     for (auto it: removable_iterators)
00306     {
00307         actuator.actions_map.erase(it);
00308     }
00309 }
00310
00311 template<typename key_t, typename action_t, typename ...Actions>
00312 actuator<action_t> connect(std::pair<key_t, action_t*> A1, Actions... An)
00313 {
00314     using actuator_t = untangle::actuator<action_t>;
00315     actuator_t actuator;
00316     actuator.actions_map = {A1, An...};
00317
00318     // remove empty actions
00319     remove_empty_actions(actuator);
00320     return actuator;
00321 }
00322
00323 // generic helpers to remove const qualifier from a function type,
00324 // for instance const member functions
00325 template <typename T>
00326 struct function_remove_const;
00327
00328 template <typename R, typename... Args>
00329 struct function_remove_const<R(Args...)>
00330 {
00331     using type = R(Args...);
00332 };
00333
00334 template <typename R, typename... Args>
00335 struct function_remove_const<R(Args...)const>
00336 {
00337     using type = R(Args...);
00338 };
00339
00343
00362 template <typename class_t, typename T, typename action_t = std::function<typename
    function_remove_const<T>::type>
00363 action_t bind(const std::shared_ptr<class_t>& obj, T class_t::* method)
00364 {
00365     return [wp = std::weak_ptr<class_t>(obj), method](auto&&... args) -> typename action_t::result_type
00366     {
00367         // lock() also keeps the object alive for the duration of the call
00368         const auto obj_ = wp.lock();
00369         if (!obj_) {
00370             //inform the actuator about dead binding
00371             throw invalid_action("bind: invalid object");
00372         }
00373         return ((*obj_).*method)(std::forward<decltype(args)>(args)...);
00374     };
00375 }
00376

```

```
00392 template <typename class_t, typename T, typename action_t = std::function<typename
      function_remove_const<T>::type>
00393 action_t bind(class_t* obj, T class_t::* method)
00394 {
00395     return [obj, method](auto&&... args) -> typename action_t::result_type
00396     {
00397         if (!obj)
00398         {
00399             //inform the actuator about dead binding
00400             throw invalid_action("bind: invalid object");
00401         }
00402         return ((obj)->*method)(std::forward<decltype(args)>(args)...);
00403     };
00404 }
00405
00406 }
```


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